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Nốt

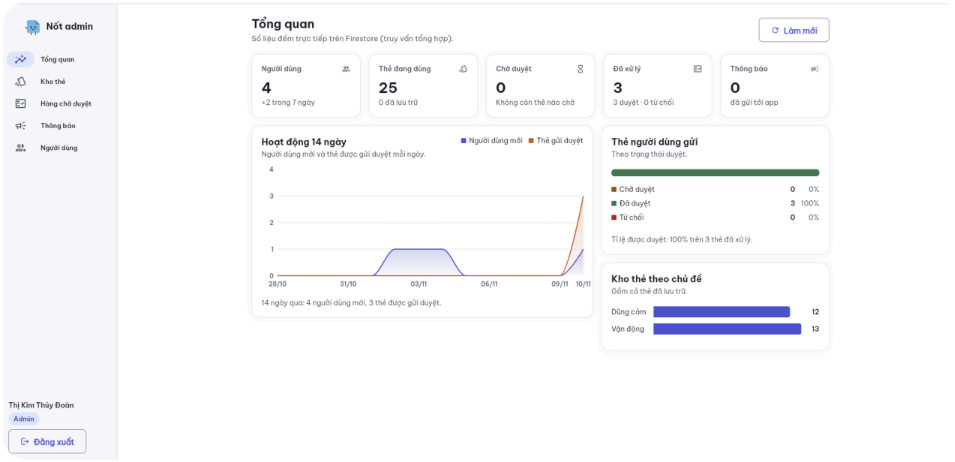
One positive note every day, now with staff, AI and a shared catalog

Đoàn Thị Kim Thúy · SE190992

Phùng Thế Quân · SE193519

Phan Đức Duy · SE193591

Phan Hữu Phước · SE161951

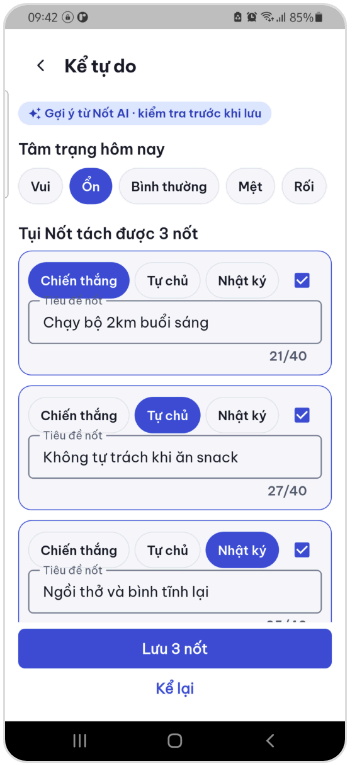




PROBLEM · 01

Overview

- Users log a win on Android
- Staff curate the cards on the web
- Gemini sorts the day into notes
- One Firestore, rules per role



PROBLEM · 02

Context

- Goal: a mood and one win in under 30 seconds
- Lab 3: one user, a fixed list of cards
- Missing: staff, an admin, a way to reach users



Weekly card, kept from Lab 3

Source: Lab 2 persona and success criterion (docs/design-traceability.md)

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Problem

Pain point	Cause	Impact
Static challenge content	Cards were a seed list; changing one needed a developer	Users run out of fresh cards
No way to reach users	No staff role, no announcements	Nobody can tell users about news in the app
Logging takes effort at night	Each note is typed and classified by hand	The 30 s goal slips on a busy day
Lab 3 flow unfinished	Weekly PDF went to Cloud Storage (paid plan)	Export never worked on the real project

Evidence: docs/gap-review.md. No user survey; impacts observed in the Lab 3 build.

PROBLEM · 04

Proposed solution

Pain point	Solution	Value
Static content	Web admin catalog; users share cards, staff approve	Catalog grows without a release
No channel	Announcements and a personal inbox	News lands in the same minute
Slow evenings	Nốt AI: tell the day once, confirm the notes	Several notes from one paragraph
Broken PDF	Build and save it on the device	Works on the free plan, offline

Out of scope: Cloud Functions, Cloud Storage, AI level 3 “Hỏi Nốt”, iOS

PROBLEM · 05

Main actors

Actor	Client	Main permissions
User	Android app	Own notes; share own cards (pending only); read catalog, announcements, own inbox
Staff	Web admin	Create, edit, archive cards; approve or reject with a reason; post announcements
Admin	Web admin	Everything staff can do, plus change other users' roles
Gemini	External	Receives the scrubbed paragraph only; returns JSON suggestions
Firebase	Platform	Rules decide every read and write; no custom server

Role lives in users/{uid}.role; only an admin changes it, never their own.

PROBLEM · 06

Main features

WF-01

Log the day with Nốt AI

User

Taps Ghi nốt → Kể tự do

→ Notes saved after confirming

WF-02

Publish a card to the catalog

User + Staff

Shares an own card

→ Card in catalog, inbox message

WF-03

Announce to all users

Staff

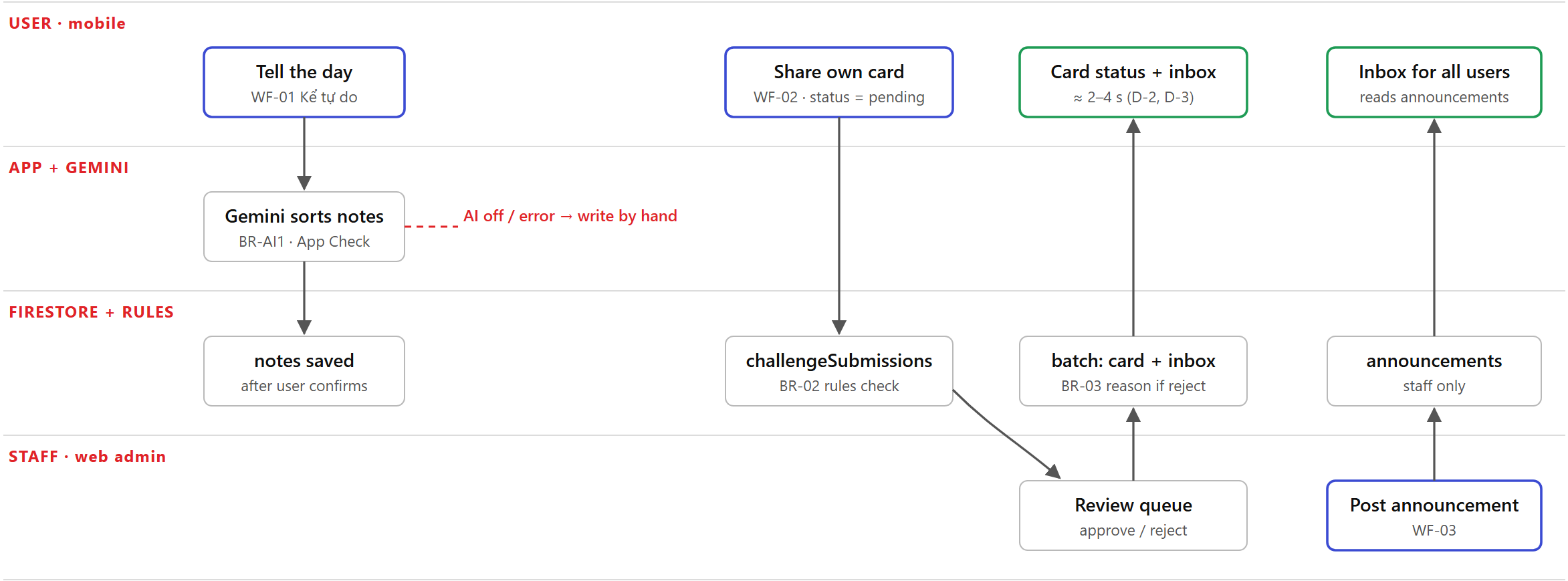
Has news for everyone

→ Shown in every user inbox

Also: onboarding · weekly PDF · streaks · self-control · groups · Discord link · dashboard

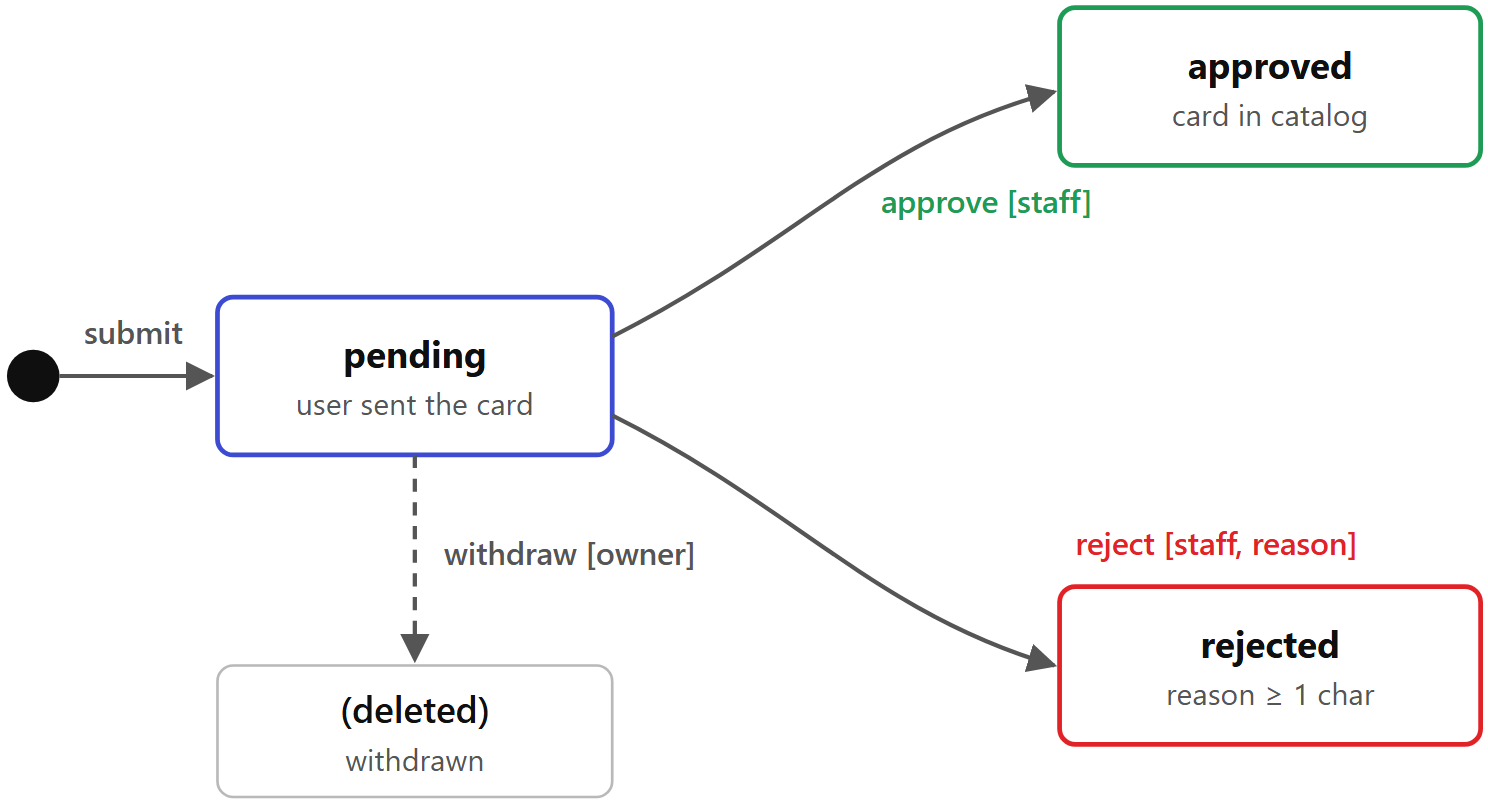
DESIGN · 07

Business flow



BR-AI1: AI only when enabled, consented, App Check · BR-02: pending only · BR-03: reject needs a reason

State machine

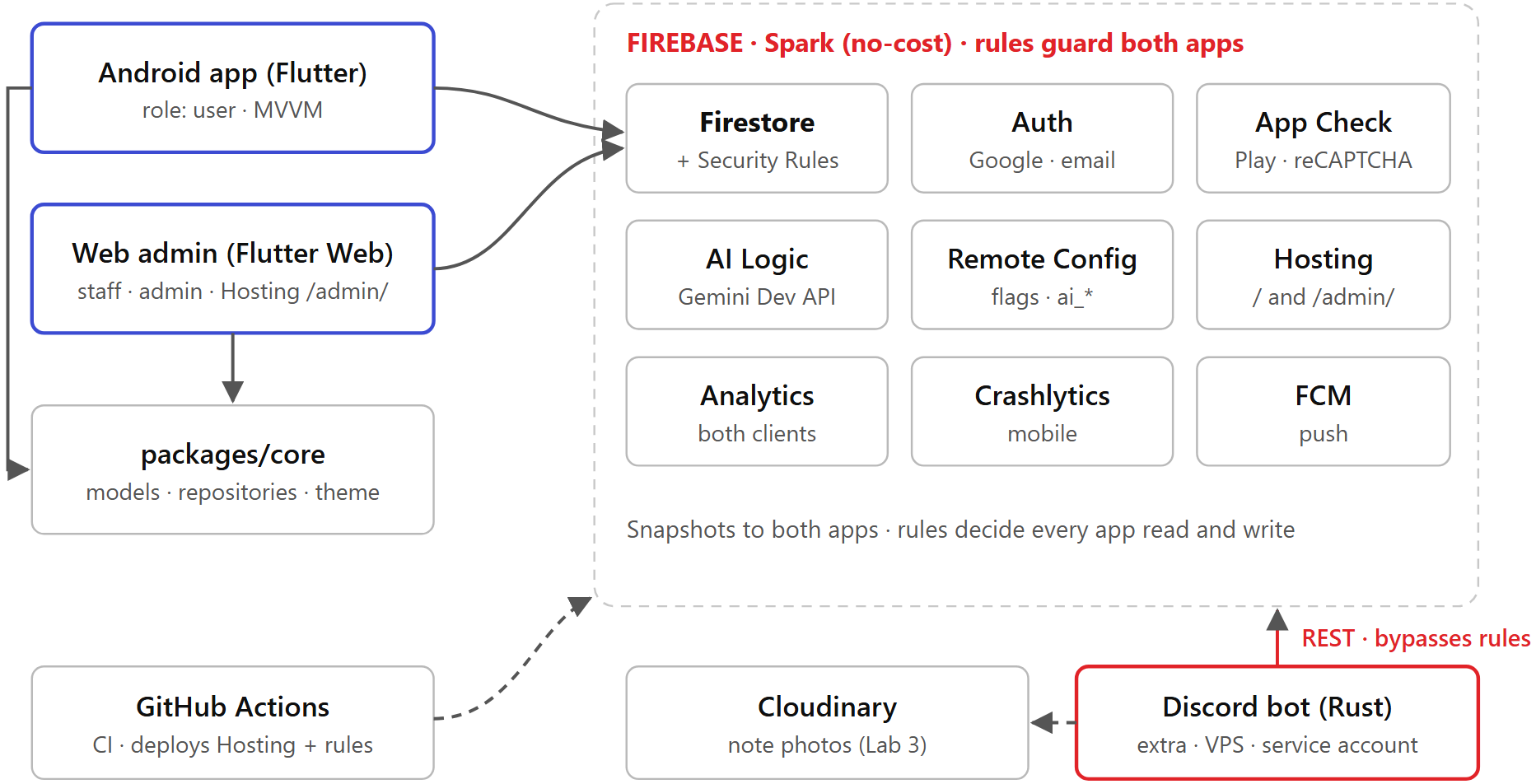


Forbidden (rules deny)

user → approved
approved → pending
reject without reason
withdraw after review

Tests R-20…R-24 · 30/30 rules tests pass

System architecture



GitHub Actions deploys Hosting (/ and /admin/) and rules on merge to main

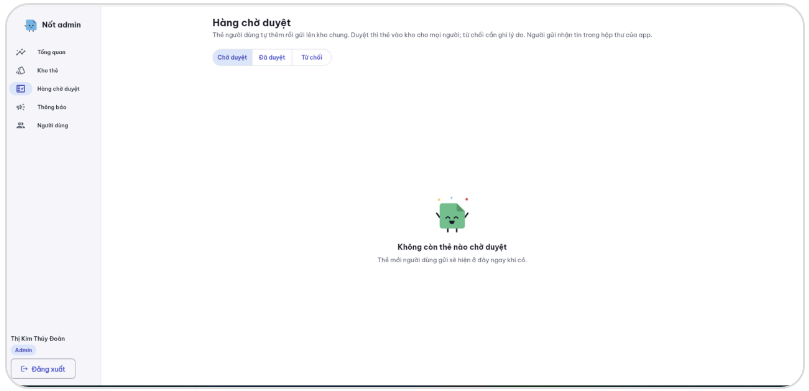
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Technology stack

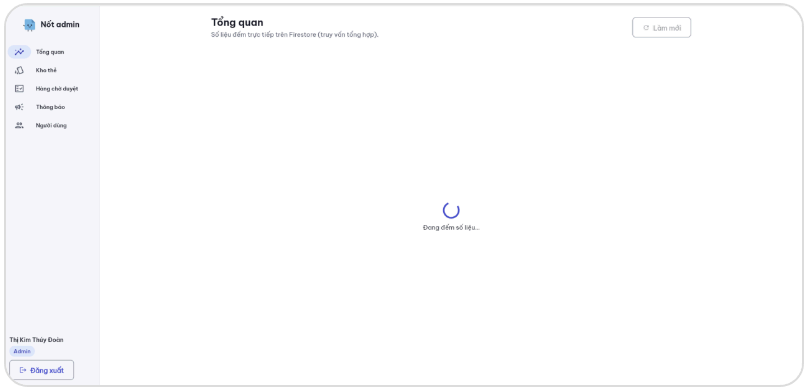
Group	Technology	Used for
Mobile	Flutter (Dart) · Riverpod · charts · PDF · sharing and camera	Android app in MVVM; charts; weekly card PDF saved and shared on the device
Web	Flutter Web (Dart) · Riverpod · Firebase Hosting	Web admin for staff and admin, served from Firebase Hosting at /admin/
Shared	Shared Dart package	Models, repositories and design tokens used by both clients
Backend	Firebase: Firestore · Rules · Auth · Remote Config · Messaging · Analytics · Crashlytics	Data, role permissions, flags, push, usage and crash reports; no custom server
AI	Firebase AI Logic · Google Gemini	Nốt AI: sort one paragraph into notes with a JSON schema, model name kept in Remote Config
Security	Firebase App Check · Google Play Integrity · reCAPTCHA Enterprise	Only our own apps reach Firestore and Gemini
Discord bot	**Rust** · Discord API · async runtime · HTTP client	Assistant in Discord DM: the connect command links a Nốt account with a 6-character code, then the user manages notes by chatting
Bot AI	LLM agent framework · OpenRouter · DeepSeek	Agent with tools bound to the linked user's own notes; a guard model and a cooldown limit abuse
Bot data	Firestore REST API · Google service account · Cloudinary	Reads and writes notes and account links; uploads photos sent in the chat
Media	Cloudinary	Note photos, because Cloud Storage needs the paid plan
Quality	Flutter tests · Firebase Emulator · ARTEMIS · CodeRabbit	Unit and widget tests, rules tests, E2E on a real phone, AI code review on PRs
DevOps	GitHub Actions · Firebase Hosting previews · Docker · Dokploy	CI, PR previews, deploy on merge; the bot ships as a Docker image on a VPS

Source: app and web admin pubspec, discord_bot/Cargo.toml, GitHub workflows

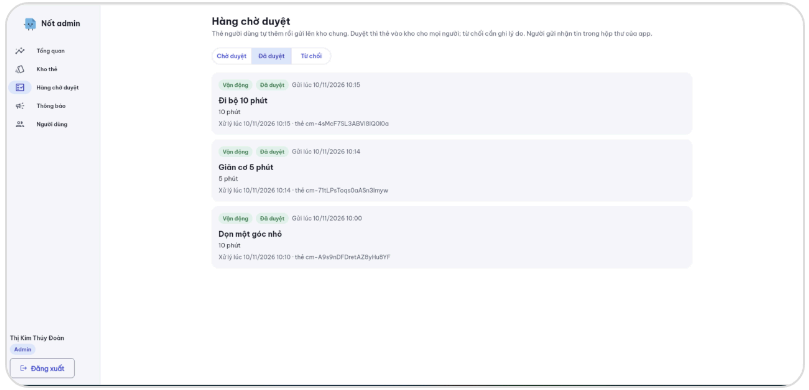
UI/UX design



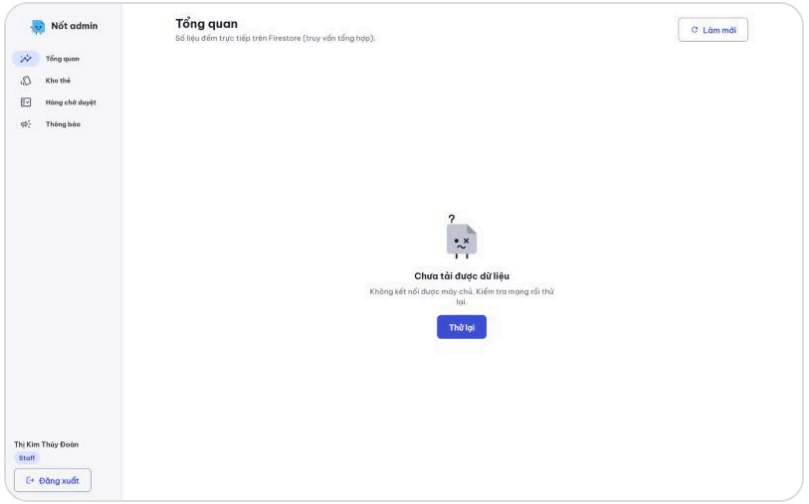
Empty: review queue



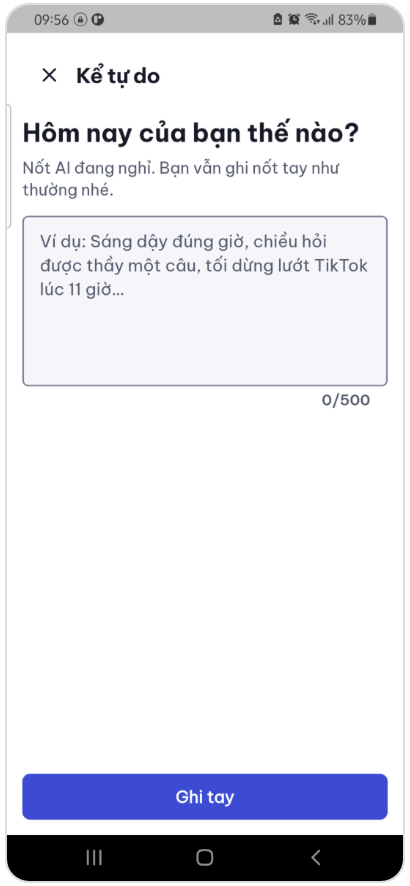
Loading: dashboard



Success: approved cards



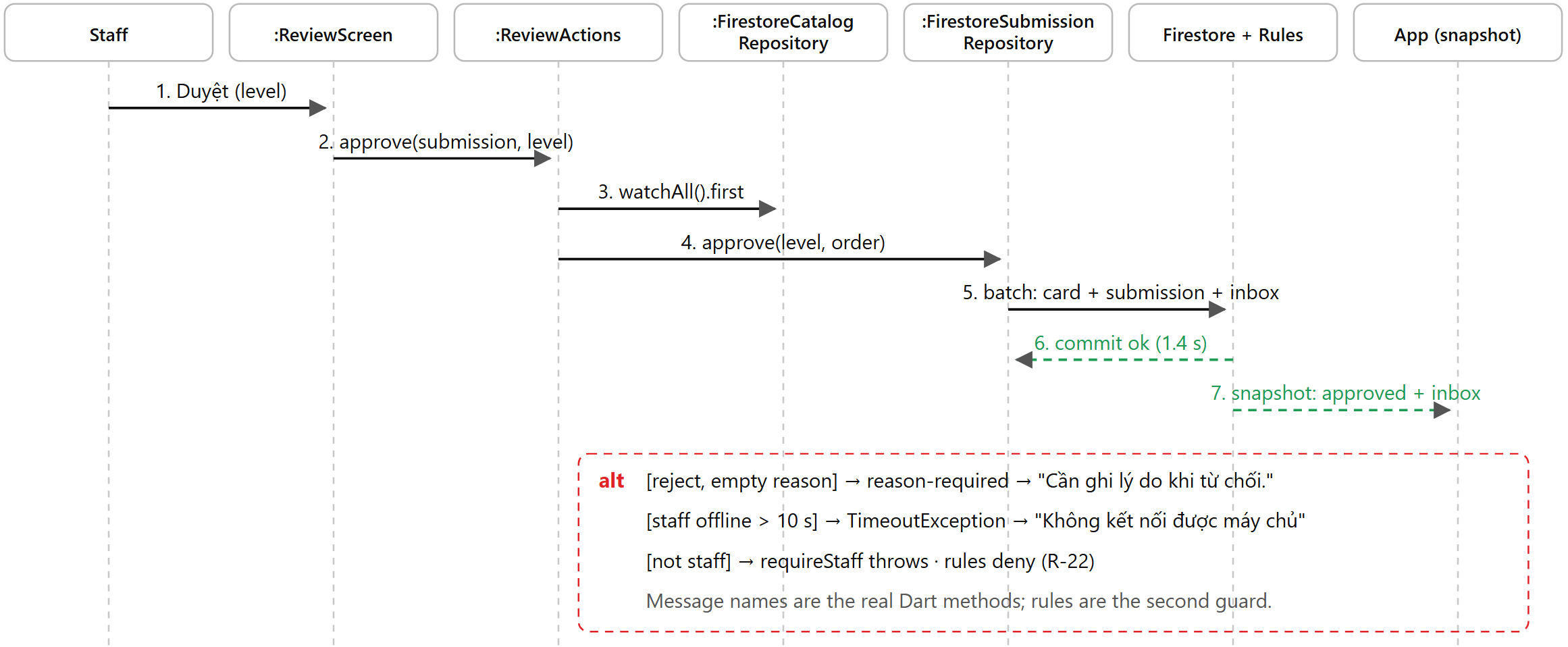
Error: dashboard, server unreachable



App: AI off

States: empty · loading · error (5 screens) · success

Sequence diagram



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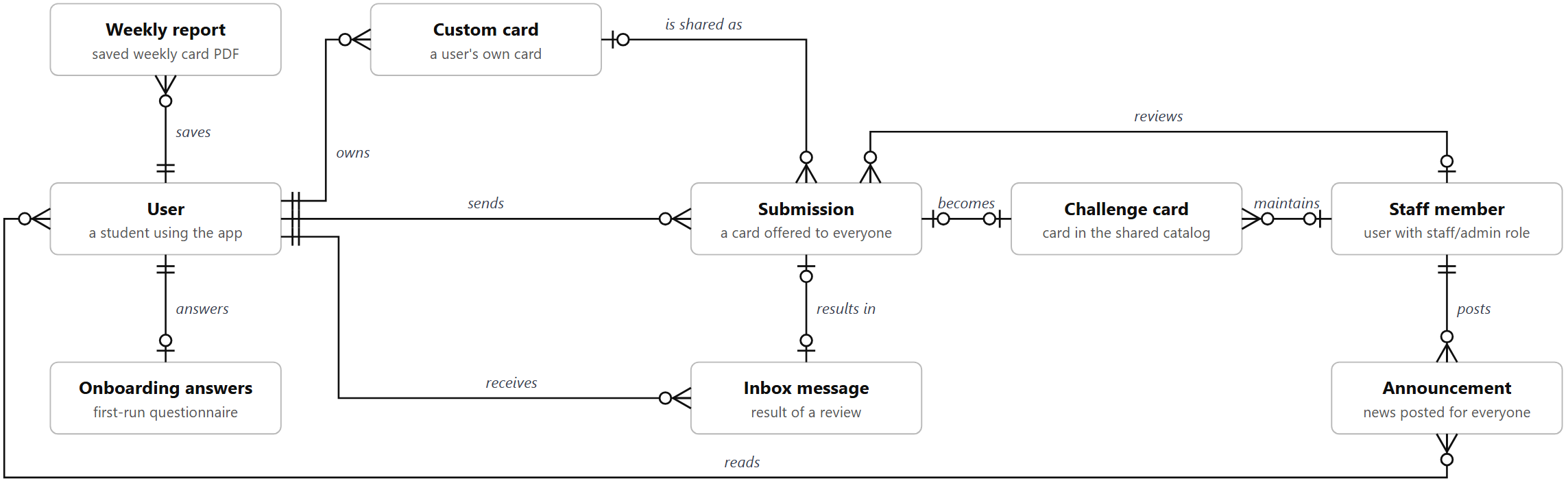
Class design

Layer	WF-01 Nốt AI	WF-02 Share & review	WF-03 Announce
Screen	`FreeTellScreen`	`ReviewScreen`	`AnnouncementsScreen`
ViewModel	`FreeTellViewModel`	`ReviewActions`	`AnnouncementActions`
Service	`AiService`	`SubmissionRepository`	`AnnouncementRepository`
Implementation	`FirebaseAiService`	`FirestoreSubmissionRepository`	`FirestoreAnnouncementRepository`
Model	`AiSortResult`	`ChallengeSubmission`	`Announcement`

Tests swap in fakes · 290 app · 10 web admin · 5 core

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Database design: conceptual

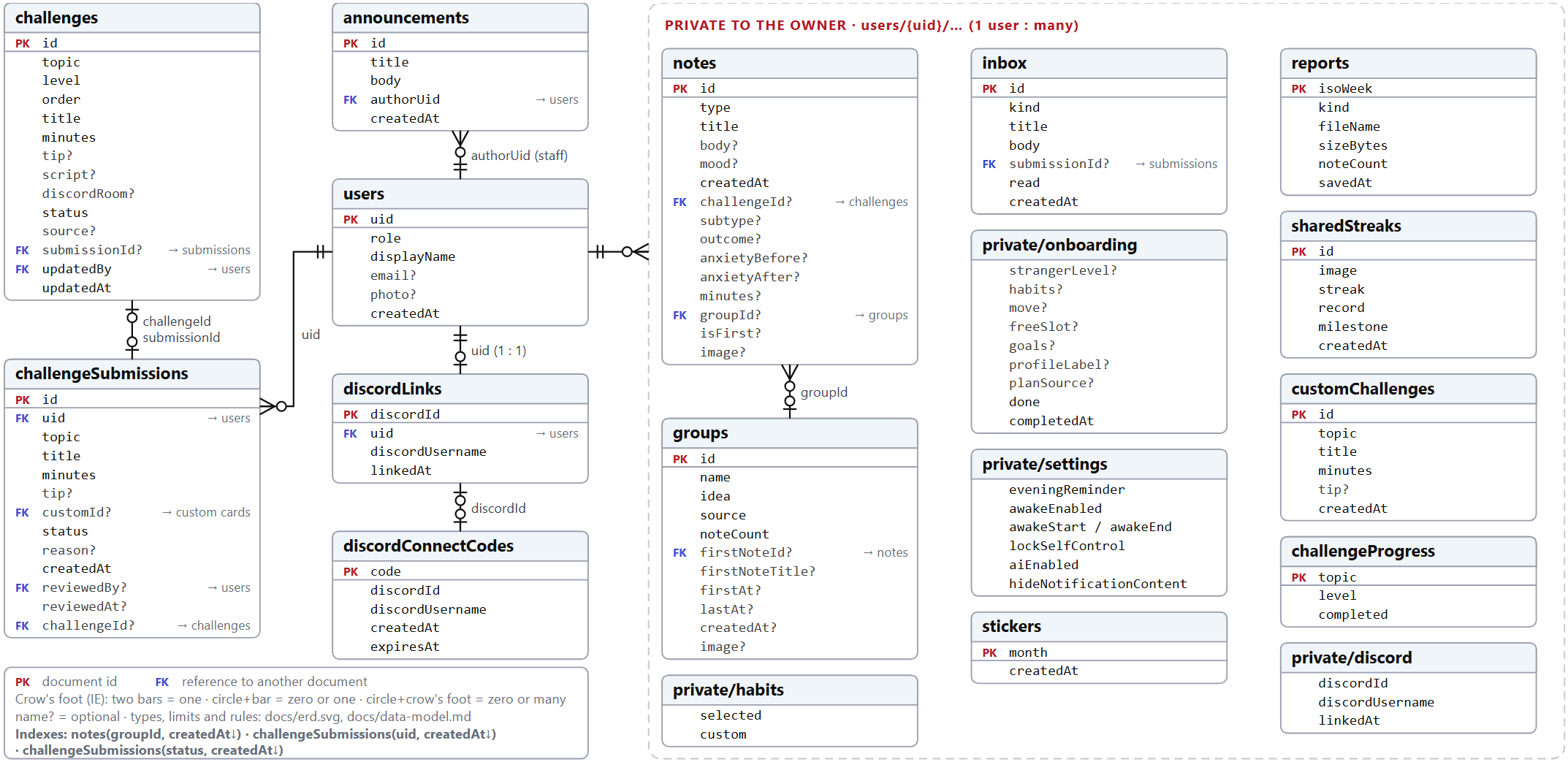


Crow's foot (IE) notation: two bars = exactly one · circle + bar = zero or one · circle + crow's foot = zero or many
Business level only: no attributes, keys or collection names (see the physical model on the next slide).

Final additions only; Lab 3 data (notes, groups, habits, …) is not drawn · attributes: next slide

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Database design: physical



Types, limits and rules per field: docs/erd.svg · docs/data-model.md

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Traceability

Workflow	Screen	Class	Collection	Tests
WF-01 Nốt AI	Kể tự do	FreeTellViewModel	notes	free_tell_flow_test · real device
WF-02 Share	Review queue	ReviewActions	challengeSubmissions	R-20…R-24 · D-1…D-3
WF-03 Announce	Announcements	AnnouncementActions	announcements	R-25 · inbox test
Weekly PDF	Thẻ tuần	WeeklyReportViewModel	reports	R-28 · real device
Onboarding	S13a–g	OnboardingViewModel	private/onboarding	R-27 · flow test
Roles	Sign-in · Users	sessionProvider	users	R-01, R-26

Full matrix: docs/design-traceability.md

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DEMO · 17

Demo plan

01 · USER

Kể tự do

One paragraph becomes 3 saved notes

02 · USER → STAFF

Share and approve

“Đi bộ 10 phút” shows up in the app

03 · STAFF → USERS

Announce

“Bạn đã vận động chưa?” in every user inbox

> Shown live: AI off, so the user writes by hand · reject without a reason is blocked

Ready: admin account · Galaxy A70 with debug token · ai_enabled toggle · backup video

DEMO · 18

WF-01 · Nốt AI

TRIGGER

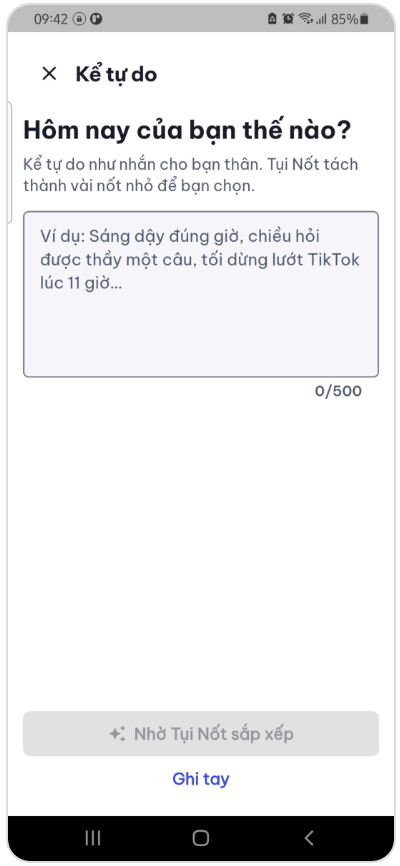
Evening: Ghi nốt → Kể tự do

RESULT

Gemini returns labelled notes; the user edits and saves

EXCEPTION

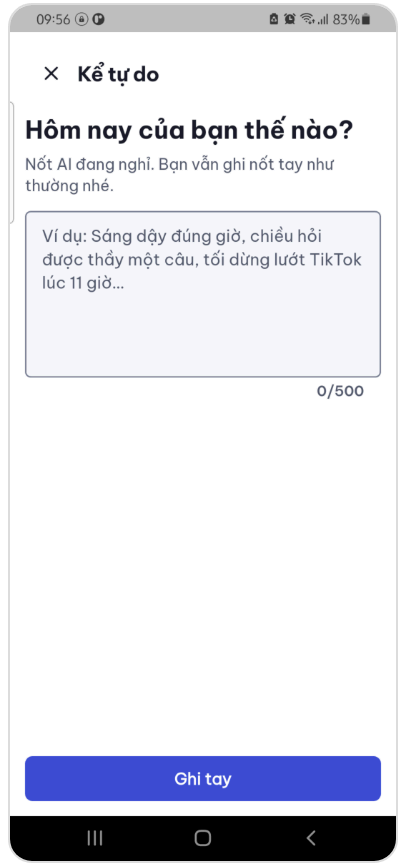
> AI off or error: message and “Ghi tay”, text kept



Input (consent sheet on first use)



Suggestions

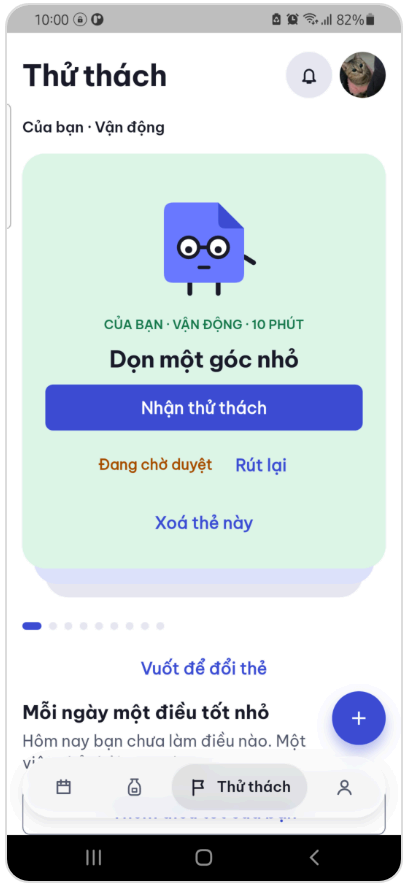


ai_enabled = false

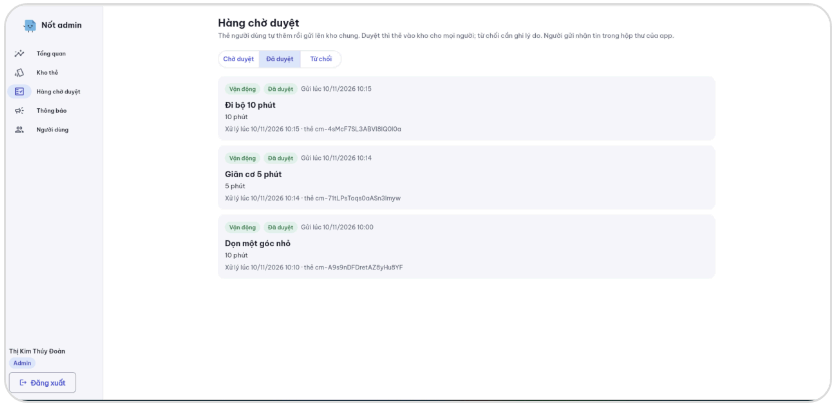
Galaxy A70 · real Gemini (~2 s) · gemini-flash-lite-latest

DEMO · 19

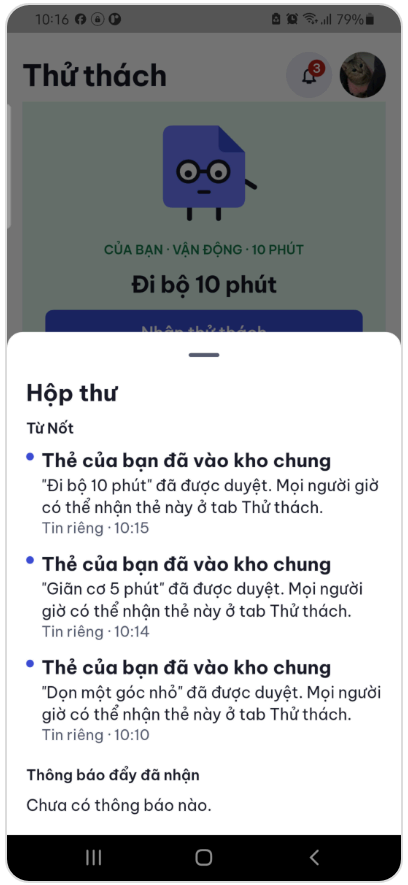
WF-02 · Share a card



1 · Đang chờ duyệt



2 · Web: approved

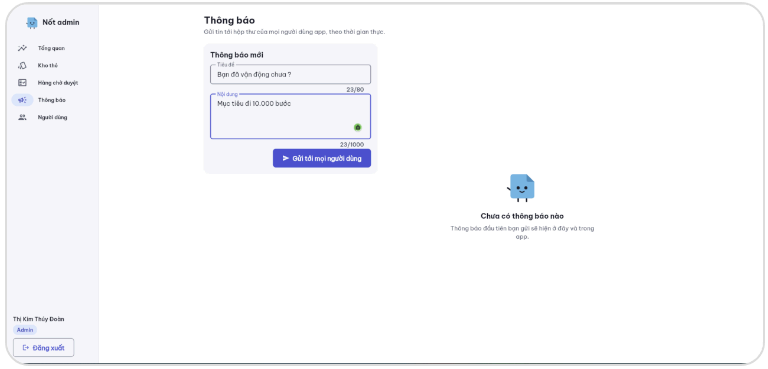


3 · App: inbox

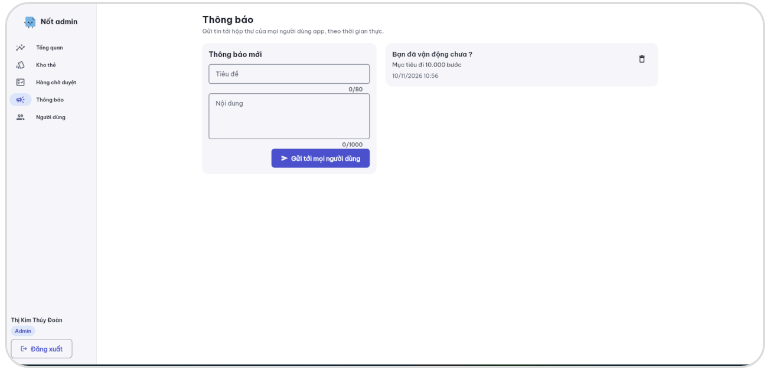
Server write 1.4 s after Duyệt · phone ≤ 3.4 s (D-2), ≤ 2.3 s (D-3), 22 s (D-1, idle phone)

DEMO · 20

WF-03 · Announce



1 · Compose



2 · Sent at 10:56



3 · App inbox, 10:56

Rules R-25: staff only · authorUid = self · title ≤ 80 · body 1–1000

PROOF · 21

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Testing

30/30
rules tests

290
app tests

10 + 5
web + core tests

41/42
ARTEMIS runs (Lab 3)

Level	Evidence	Result
Rules	Emulator: allowed and denied per role per collection	30 / 30
Unit / widget	ViewModels, repositories, screens at 360 dp	all pass
Cross-role	3 timed runs on the deployed system	3 / 3
Web checklist	13 screenshots: actions, rejection, empty, loading	done
Real device	Onboarding, Nốt AI, PDF, AI off on Galaxy A70	pass
E2E (ARTEMIS)	Lab 3 flows 3× each; Final flows written, not run	partial

CI on every PR: format, analyze, all suites, rules in the Emulator, builds, Hosting preview

PROOF · 22

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Achievements

Brief requirement	Implemented	Evidence
Web admin, deployed	Dashboard, catalog, review, announcements, users	`evidence/web-admin/`
Role-based rules, tested	user / staff / admin, R-01…R-30	`evidence/firestore/`
Cross-role flow, real time	Share card → approve → app + inbox	`docs/cross-role-flow.md`
AI level 2 with fallback	Kể tự do on Gemini, Remote Config flags	`docs/ai-feature.md`
App Check	reCAPTCHA Enterprise · Play Integrity	`evidence/app-check/`
Lab 3 PDF on device	pdf + share_plus, metadata in Firestore	15.4 KB file from the phone
Mobile data with inbox	Announcements + personal inbox	web checklist #11

PROOF · 23

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Limitations & next steps

Limitation	Impact	Priority	Next step
ARTEMIS not run for Final flows	No 3× E2E evidence for them	High	Run on the Galaxy A70, 3× each
App Check on Authentication only monitored	26% of sign-in requests are unverified	Medium	Enforce once all traffic is verified
Lab 2 flows A3 and P4 not built	Two designed flows missing	Medium	Build A3 on the same AiService
D-1 took 22 s on an idle phone	Real time not guaranteed after idle	Low	Re-listen on resume
Sideloaded release builds fail Play Integrity	AI blocked outside Play / debug	Low	Demo on debug; Play internal track

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Lessons learned

Approve timed out

Riverpod paused the unwatched provider. Fixed: read once (PR #70).

Dashboard spun offline

`count()` waits forever. Fixed: 15 s timeout (PR #71).

Gemini model retired

404 for new projects. Fixed: model name in Remote Config.

Rules are the backend

Every business rule is a rules condition, so rules need tests.

Read the brief twice

We removed the PDF, then re-read: the brief wanted it on the device.

AI needs a plan B

App Check, quota and model changes broke AI; hand-writing kept the flow alive.

PROOF · 25

Conclusion

1

Problem solved

Staff curate the cards; news reaches every user in the same minute

2

Working product

Three workflows run live on Android and the web admin

3

Verified

30/30 rules tests, 3 timed runs, real-device evidence

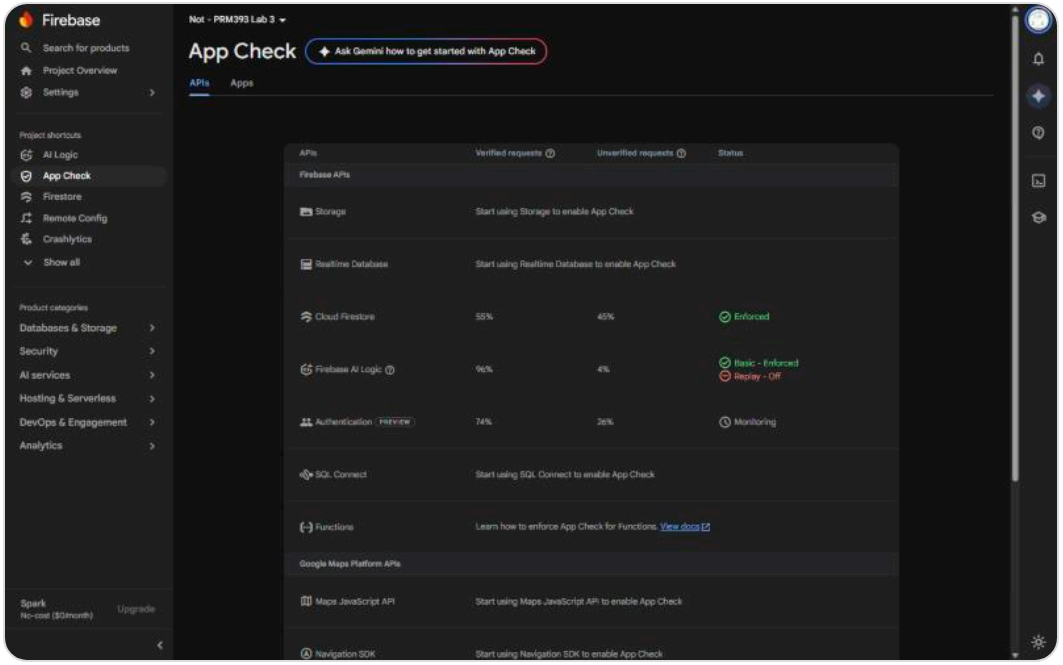
APPENDIX · A1

AI examples

Input (Kể tự do)	Gemini output
Ran 2 km, ate a snack I meant to skip, calmed down without blaming myself	chien_thang “Chạy bộ 2km quanh hồ” · nhat_ky · tu_chu · mood binh_thuong
Asked a stranger in the library where to charge my phone; wrote a few lines about the rainy afternoon	chien_thang “Lấy hết can đảm hỏi mượn chỗ sạc pin” · nhat_ky · mood vui
Tired for days, only tidied my desk today	chien_thang “Mình đã dọn dẹp được cái bàn học” · 2× nhat_ky · mood met
Handled failure	404 or 401 → `ai_error` logged, text kept, “Ghi tay” offered

Inputs are scrubbed (email, phone) and capped at 500 characters

App Check



Cloud Firestore and AI Logic: Enforced



Galaxy A70, 5 min after enforcing: Hũ loads

Web: reCAPTCHA Enterprise · Android: Play Integrity + debug token · evidence/app-check/ (12/11/2026)